

Square clocks, run recurrences, and an Euler-product floor for Möbius distance-two capacity

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Abstract

The RH-373 composite clock gives one strict improvement to the unconditional lower floor for the RH-366 distance-two Möbius capacity. We construct and optimize a whole square-clock family. If $3 = p_1 < p_2 < \dots$ are the odd primes, $P_y = \prod_{i \leq y} p_i^2$, $q_y = 4P_y$, and $A_y = \prod_{i \leq y} (p_i^2 - 1)$, then the exact optimum among universally safe one-site phase/current-input factors at the fixed clock q_y is

$$B_y = \frac{4 + 2O_y/A_y}{\pi^2}.$$

Here O_y counts odd-length positive runs in one period of a squarefree word, and it has a finite Euler-product formula. Adjoining a prime $p \geq 5$ gives $A' = (p^2 - 1)A$ and $O' = (p^2 - 1)O + L_{\text{even}}$, where L_{even} counts sites in even-length old runs. A length-eight run persists by the Chinese remainder theorem, so B_y increases strictly. The fixed-clock certificates converge to an explicit Euler-product constant B_∞ , and

$$\liminf_{N \rightarrow \infty} \frac{K_N}{N} \geq B_\infty.$$

The quantifier fixes y before $N \rightarrow \infty$; no growing clock, uniform Davenport theorem, or infinite selector is asserted. The optimization does not include arbitrary memory transducers or all clocks, and the adaptive capacity limit, intrinsic operators, traces, Riemann-zero identification, and RH remain open or outside scope.

1 Frozen capacity and the exact selector class

The RH-366 distance-two capacity is

$$K_N = \max_{\substack{\varepsilon_1, \dots, \varepsilon_N \in \{-1, +1\} \\ \text{not } (\varepsilon_n, \varepsilon_{n+2}) = (+1, +1)}} \left| \sum_{n \leq N} \mu(n) \varepsilon_n \right|. \quad (1)$$

Its optimizer may read the full finite prefix and may change with N . RH-366 proves

$$\frac{4}{\pi^2} \leq \liminf_{N \rightarrow \infty} \frac{K_N}{N} \leq \limsup_{N \rightarrow \infty} \frac{K_N}{N} \leq \frac{6}{\pi^2}, \quad (2)$$

without proving an ordinary limit [6]. RH-371 isolates an exact eight-run finite-prefix ledger, but the required Möbius run frequencies remain open [5]. RH-373 instead supplies the externally prescribed $q = 180$ certificate $97/(24\pi^2)$ [4].

We now freeze the narrower class that is optimized here.

Definition 1.1 (Universally safe one-site factor). For a fixed clock q , a one-site phase/current-input factor is a family

$$g_r : \{-1, 0, +1\} \longrightarrow \{-1, +1\}, \quad r \in \mathbb{Z}/q\mathbb{Z}.$$

It is *universally safe* when, for every ternary input word a_n , the word $g_{n \bmod q}(a_n)$ has no two $+1$ values at distance two. Its active phase set is

$$S_g = \{r : g_r(a) = +1 \text{ for at least one } a \in \{-1, 0, +1\}\}.$$

Universal safety is equivalent to $S_g \cap (S_g + 2) = \emptyset$. Necessity follows by choosing the two activating input values independently; sufficiency is immediate. This is the one-site factor boundary used for finite transducer certificates in RH-372 [3]. It is not the class of all finite-memory transducers.

For $r \pmod{q}$, let

$$\delta_{q,r} = \lim_{N \rightarrow \infty} \frac{1}{N} \#\{n \leq N : n \equiv r \pmod{q}, \mu(n)^2 = 1\}.$$

Fixed-frequency Davenport cancellation, followed by finite Fourier inversion, gives

$$M_r(N) := \sum_{\substack{n \leq N \\ n \equiv r \pmod{q}}} \mu(n) = o(N)$$

for every fixed q, r . The squarefree progression sieve supplies $\delta_{q,r}$ [1, 2]. Splitting the squarefree terms according to $\mu(n) = \pm 1$ therefore gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} \mu(n) g_{n \bmod q}(\mu(n)) = \frac{1}{2} \sum_{r \bmod q} \delta_{q,r} (g_r(+1) - g_r(-1)). \quad (3)$$

For an active phase, the absolute contribution in (3) is at most $\delta_{q,r}$; an inactive phase contributes zero. Conversely, any distance-two independent set I realizes its full positive weight by taking

$$g_r(+1) = +1, \quad g_r(0) = g_r(-1) = -1 \quad (r \in I), \quad g_r \equiv -1 \quad (r \notin I). \quad (4)$$

Thus the maximum absolute limiting correlation in this fixed- q class is exactly the maximum-weight independent-set value for the phase graph, with vertex weights $\delta_{q,r}$.

2 Square clocks and the run formula

Let

$$3 = p_1 < p_2 < \cdots, \quad P_y = \prod_{i \leq y} p_i^2, \quad q_y = 4P_y, \quad A_y = \prod_{i \leq y} (p_i^2 - 1), \quad (5)$$

and, for $1 \leq m \leq 9$, put

$$E_m^{(y)} = \prod_{i \leq y} \left(1 - \frac{m}{p_i^2}\right). \quad (6)$$

Lemma 2.1 (Equal positive progression weights). *For $q = q_y$, a squarefree progression density is zero exactly when $4 \mid r$ or $p_i^2 \mid r$ for some $i \leq y$. Every other phase has*

$$\delta_{q_y, r} = \frac{2}{A_y \pi^2}. \quad (7)$$

Proof. The zero cases force a square divisor. In every other class the fixed progression sieve gives

$$\frac{1}{q_y} \prod_{p \nmid q_y} (1 - p^{-2}) = \frac{1}{4P_y} \frac{6/\pi^2}{(1 - 2^{-2}) \prod_{i \leq y} (1 - p_i^{-2})} = \frac{2}{A_y \pi^2}.$$

□

The distance-two phase graph has an even and an odd cycle, each of length $2P_y$. On the even cycle, every positive-weight phase is $2 \pmod{4}$. There are A_y such phases by the Chinese remainder theorem, and no two are adjacent in the distance-two graph. Hence the even contribution to the maximum independent-set cardinality is exactly A_y .

Parametrize the odd phases by $r = 2k + 1$. Their positive-weight indicator is the P_y -periodic word

$$w_y(k) = \mathbf{1}_{\{p_i^2 \nmid 2k+1 \text{ for every } i \leq y\}}. \quad (8)$$

One period has A_y one-sites. Because $p_1^2 = 9$, a zero occurs every nine positions, so every positive run has length at most eight. Let $R_j^{(y)}$ be the number of cyclic runs of exact length j in one period and define

$$O_y = R_1^{(y)} + R_3^{(y)} + R_5^{(y)} + R_7^{(y)}.$$

Lemma 2.2 (Exact run ledger). *For $1 \leq j \leq 7$,*

$$R_j^{(y)} = P_y \left(E_j^{(y)} - 2E_{j+1}^{(y)} + E_{j+2}^{(y)} \right), \quad (9)$$

whereas the terminal row is separately

$$R_8^{(y)} = P_y E_8^{(y)}. \quad (10)$$

In particular,

$$O_y = P_y \sum_{j \in \{1,3,5,7\}} \left(E_j^{(y)} - 2E_{j+1}^{(y)} + E_{j+2}^{(y)} \right). \quad (11)$$

Proof. Let N_m count cyclic starting positions at which m consecutive entries of w_y equal one. For $m \leq 9$, the m forbidden congruence classes modulo each p_i^2 are distinct. The Chinese remainder theorem gives

$$N_m = P_y E_m^{(y)}.$$

For $j \leq 7$, inclusion–exclusion on extension by one site to the left and right gives $R_j = N_j - 2N_{j+1} + N_{j+2}$. The factor at $p_1 = 3$ makes $E_9^{(y)} = 0$, hence $N_9 = 0$ and no run exceeds eight. Therefore a length-eight block is already a whole run, so $R_8 = N_8$; no E_{10} term is introduced. Summing the four odd rows proves (11). $\square$

Theorem 2.3 (Fixed square-clock optimum). *Among universally safe one-site phase/current-input factors at the fixed clock q_y , the exact maximum absolute limiting correlation is*

$$B_y = \frac{4 + 2O_y/A_y}{\pi^2}. \quad (12)$$

Consequently

$$B_y \leq \liminf_{N \rightarrow \infty} \frac{K_N}{N}. \quad (13)$$

Proof. The odd phase cycle contains two copies of the P_y -periodic word. A run of length ℓ contributes $\lceil \ell/2 \rceil$ vertices to a maximum independent set. Hence the two copies contribute

$$2 \sum_{\text{runs}} \left\lceil \frac{\ell}{2} \right\rceil = \sum_{\text{runs}} (\ell + \mathbf{1}_{\{\ell \text{ odd}\}}) = A_y + O_y.$$

Adding the A_y even phases gives $2A_y + O_y$ positive phases. Every one has weight $2/(A_y \pi^2)$ by (7), so the phase MWIS value is (12). Equation (4) realizes that value, and its output is admissible in (1) for every N . $\square$

Remark 2.4 (Exact scope of “optimum”). The theorem optimizes one fixed q_y and only the universally safe one-site phase/current-input factor class. It is not an optimum over arbitrary memory transducers, over all clocks, or over the adaptive finite-prefix optimizers defining K_N .

3 Prime adjoining and strict monotonicity

Adjoin a new odd prime $p \geq 5$ to an old clock, and write A', O' for the new quantities. Let

$$L_{\text{even}} = \sum_{\substack{\text{old runs } R \\ |R| \text{ even}}} |R| \quad (14)$$

be the number of old one-sites contained in even-length runs.

Proposition 3.1 (Run recurrence). *For every newly adjoined $p \geq 5$,*

$$A' = (p^2 - 1)A, \quad O' = (p^2 - 1)O + L_{\text{even}}. \quad (15)$$

Proof. Before imposing the new condition $p^2 \nmid 2k + 1$, one new period consists of p^2 copies of the old cyclic word. Every old one-site has p^2 lifts, and exactly one lift is deleted, proving the formula for A' . Within a fixed old run of length $\ell \leq 8$, its ℓ deleted lifts occur in distinct copies: two positions assigned to the same copy would differ by zero modulo p^2 , impossible because $0 < |j - j'| \leq 7 < p^2$.

Suppose first that ℓ is odd. There are $p^2 - \ell$ untouched copies of the odd run. Deleting position $t \in \{0, \dots, \ell - 1\}$ leaves runs of length t and $\ell - 1 - t$. The $(\ell + 1)/2$ even values of t produce no odd subrun, while the $(\ell - 1)/2$ odd values produce two. The total number of odd runs descending from this old run is therefore

$$(p^2 - \ell) + 2 \frac{\ell - 1}{2} = p^2 - 1.$$

If ℓ is even, each of its ℓ deletions splits the run into one odd and one even subrun, contributing ℓ odd runs in total. Summing over old odd and even runs proves (15). $\square$

Lemma 3.2 (Persistent length-eight run). *For every $y \geq 1$, the cyclic word w_y has a run of exact length eight.*

Proof. Choose $k \equiv 5 \pmod{9}$. Then the two boundary values at $k - 1$ and $k + 8$ are zero modulo 9, while the eight values $2(k + j) + 1$, $0 \leq j \leq 7$, are nonzero modulo 9. For each other prime square $p_i^2 \geq 25$, choose $k \pmod{p_i^2}$ outside the eight forbidden classes $2(k + j) + 1 \equiv 0 \pmod{p_i^2}$. Such a choice exists because $p_i^2 > 8$. The Chinese remainder theorem joins these choices. The eight interior sites are then one-sites for every old prime square, and the two boundary sites are zeros because of 9. $\square$

The lemma gives $L_{\text{even}} \geq 8$. Dividing (15) by $A' = (p^2 - 1)A$ yields

$$\frac{O'}{A'} = \frac{O}{A} + \frac{L_{\text{even}}}{(p^2 - 1)A} > \frac{O}{A}. \quad (16)$$

Thus $B_{y+1} > B_y$ at every step.

4 Euler limit and the capacity floor

For $1 \leq m \leq 9$, define

$$e_m = \prod_{p \text{ odd}} \left(1 - \frac{m}{p^2}\right). \quad (17)$$

The products are positive for $m \leq 8$ because $\sum_p p^{-2} < \infty$, while $e_9 = 0$ because the $p = 3$ factor vanishes. Moreover

$$e_1 = \prod_{p \text{ odd}} (1 - p^{-2}) = \frac{8}{\pi^2}. \quad (18)$$

Put

$$C = \frac{1}{e_1} \sum_{j \in \{1,3,5,7\}} (e_j - 2e_{j+1} + e_{j+2}), \quad B_\infty = \frac{4 + 2C}{\pi^2}. \quad (19)$$

Theorem 4.1 (Euler-product capacity floor). *The sequence B_y is strictly increasing and converges to B_∞ . Furthermore,*

$$\liminf_{N \rightarrow \infty} \frac{K_N}{N} \geq B_\infty. \quad (20)$$

Proof. Since $A_y = P_y E_1^{(y)}$, equations (11) and (12) give

$$\frac{O_y}{A_y} = \frac{\sum_{j \in \{1,3,5,7\}} (E_j^{(y)} - 2E_{j+1}^{(y)} + E_{j+2}^{(y)})}{E_1^{(y)}}.$$

Termwise convergence of the finitely many Euler products proves $B_y \rightarrow B_\infty$; strictness was proved in (16).

For the last assertion, fix y first. The clock q_y is then finite, so the fixed-frequency arithmetic used in (3) proves $\liminf K_N/N \geq B_y$. The same scalar liminf dominates every B_y . It therefore dominates $\sup_y B_y = B_\infty$. This argument neither chooses $y = y(N)$ nor requires a Davenport estimate uniform in q_y . $\square$

Remark 4.2 (No infinite selector). Equation (20) takes the supremum of already proved scalar lower bounds. It does not construct a selector with an infinite clock and does not exchange the y and N limits.

5 Exact rows and executable audit

The first four exact rows are

y	q_y	A_y	O_y	L_{even}	selected	$\pi^2 B_y$
1	36	8	0	8	16	4
2	900	192	8	160	392	49/12
3	44100	9216	544	7160	18976	593/144
4	5336100	1105920	72440	838240	2284280	57107/13824

In particular,

$$\frac{49}{12} - \frac{97}{24} = \frac{1}{24}, \quad (21)$$

so the fixed $q = 900$ selector improves the RH-373 coefficient by $1/(24\pi^2)$.

The artifact evaluates the Fraction formulas through $y = 6$, directly builds the cyclic words and phase MWIS only for $y = 1, 2, 3$, and checks the recurrence for $3 \rightarrow 5$, $5 \rightarrow 7$, and $7 \rightarrow 11$. At $q = 900$ it constructs an explicit 392-phase selector, finds zero cyclic distance-two conflicts, and checks all $900 \cdot 3^2 = 8100$ phase/input-pair safety rows. The sorted selector has SHA-256

`a8e41ecc18c1cb6dec3e5aaeb64868beba8a11c095e65731c48f1a700a10053a.`

A direct finite Möbius check at $N = 2^{16}$ gives selector score 27174 and exact open-prefix capacity 32320; all 2048 initial prefixes satisfy the witness inequality. These rows are non-asymptotic reproduction checks. They do not prove (20).

For orientation only, a prime product through 200000 plus the elementary tail estimate

$$-\log \prod_{p > X} (1 - m/p^2) \leq \frac{m}{X(1 - m/X^2)} \quad (1 \leq m \leq 8)$$

gives the conservative enclosure

$$4.164146694132415320 < \pi^2 B_\infty < 4.164353662045959253,$$

or

$$0.421916271909824186 < B_\infty < 0.421937242143800443.$$

The enclosure is explicitly labelled a numerical diagnostic; the theorem is the exact Euler expression (19).

6 Route verdict and Gate ledger

Route A is **G0**: the fixed-clock optimization, run recurrence, strict monotonicity, Euler limit, and scalar-supremum argument are unconditional. Route B is **STOP_SCOPED**. The result does not optimize arbitrary memory transducers or all clocks, does not supply a growing-clock theorem, and does not prove convergence of the adaptive capacity K_N/N . A theorem for the RH-371 run envelope or higher-order Möbius correlations would be a different mathematical input.

Gates A–E remain false/open. The selectors use externally prescribed arithmetic phase and the current Möbius input. They supply no canonical dynamical operator, Fredholm determinant, scattering completion, or self-adjoint generator. They also supply no intrinsic $T \log T$ law, von-Mangoldt trace, completed-zeta divisor equality, Riemann-zero identification, Hilbert–Polya construction, or proof of RH.

7 Conclusion

Square clocks turn the RH-373 example into a rigorous family theorem. The finite combinatorics are exact: equal positive progression weights reduce the fixed-clock selector class to a cyclic MWIS, the run ledger has a closed Euler-product form, and adjoining a prime creates a positive correction from even-length runs. The supremum of these fixed certificates is therefore an explicit Euler-product lower floor. The next route must address a genuinely larger selector class or the adaptive run envelope, rather than promote this family optimum beyond its proved scope.

References

- [1] Harold Davenport. On some infinite series involving arithmetical functions (II). *The Quarterly Journal of Mathematics*, os-8(1):313–320, 1937.
- [2] Leon Mirsky. Arithmetical pattern problems relating to divisibility by r th powers. *Proceedings of the London Mathematical Society*, s2-50(1):497–508, 1948.
- [3] RH research program. Bounded constraint-graph transducer certificates, 2026. Repository paper RH-372, frozen release.
- [4] RH research program. Composite-clock phase selectors and an improved mobius capacity floor, 2026. Repository paper RH-373, frozen release.
- [5] RH research program. Eight-run reduction and the cyclic pair-ledger obstruction, 2026. Repository paper RH-371, frozen release.
- [6] RH research program. Möbius orthogonality, adaptive encoding, and parry covariance, 2026. Repository paper RH-366, frozen release.